NWANKWO Udoka

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EDUCATION

The Hong Kong Polytechnic University (PolyU), Hong Kong, SAR, China
Ph.D., Applied Physics, *GPA: 4.10/4.30* (2023)
Thesis: Long-range Coulomb Interaction for Reactive Atomistic Simulations: Application to Graphene-Water Capacitors
Advisor: Prof. C.H. Lam

The African University of Science and Technology (AUST), Abuja, Nigeria
M. Sc., Theoretical Physics, *CGPA: 3.73/4.00* ***(Distinction)*** (2014)
Thesis: Development of a correction term to the kinetic energy density functional
Advisor: Prof. O. Akin-Ojo

University of Nigeria Nsukka (UNN), Enugu State, Nigeria
B.Sc., Physics & Astronomy, *CGPA: 4.51/5.00* ***(First Class Honors)*** (2011)
Thesis: Synthesis of silver/silver oxide thin film in silver ion triethanolamine solution via Chemical Bath Deposition Method
Advisor: Prof. B. A. Ezeokoye

RESEARCH EXPERIENCE

Ph.D. Research, Department of Applied Physics, PolyU, Hong Kong, SAR, China

- Develop a long-range Coulomb charge estimation method called QEq long-range (QEqLR = LrQEq) for partial atomic charge computation using the Ewald summation, with in python programming language.
- Implement QEqLR and shield long-range Coulomb for ReaxFF in an MD simulator: large-scale atomic/molecular massively parallel simulator (LAMMPS)
- Combine the reformulated ReaxFF with QEqLR to study electrochemical processes in graphene-water capacitors using LAMMPS.

M.Sc. Research, Department of Theoretical Physics, AUST, Abuja, Nigeria

- Reviewed the kinetic energy density functional of Thomas Fermi and von-Weizsacker and identified disagreement with experimental results.
- Developed a correction term for the kinetic energy density functional that depends on electron density and its gradients.
- Computed corrections for the kinetic energy of some closed shell atoms under restricted Hartree-Fork scheme.

RESEARCH SKILLS

- Good analytical skills and communication proficiency in spoken and written English.
- Strong background in computational material science and molecular dynamics simulations using LAMMPS, and condensed matter codes: Quantum Expresso, VASP and PyQuante.
- Programming skills in Python, C++, and Fortran and works in Windows and Unix/Linux OS
- Uses Microsoft operating system for typing and Power Points. Graphic tools: Gnuplot, and Python multiplot packages and MS excel.
- Structures visualization tools: OVITO, VMD, AVOGADRO, VESTA
- Currently developing skills in machine learning.

PUBLICATIONS

1. **Nwankwo, U**., Wang, Y.-D., Lam, C.-H., & Onofrio, N. (2023). Charge equilibration model with shielded long-range Coulomb for reactive molecular dynamics simulations. *The Journal of Chemical Physical, 159(4).*
2. **Nwankwo, U**., Lam, C.-H., & Onofrio, N. (2023). Reactive force field potential with shielded long-range Coulomb interaction: Application to graphene-water capacitors. *The Journal of Applied Physical 138(18).*

AWARDS

- The Hong Kong Polytechnic University Research Postgraduate Studentship Award, 2018 – 2021.

- Department of Applied Physics, The Hong Kong Polytechnic University Merit Presentation Award, 2019.
- AUST/African Capacity Building Foundation (ACBF) 2013/2014 M.Sc. scholarship award.
- Senator Chukwumeriji's scholarship award for outstanding Abia North undergraduate students, 2008 – 2011.

CONFERENCE PRESENTATIONS/PARTICIPATIONS

1. "Charge Equilibration Model with Shielded Long-Range Coulomb for Reactive Molecular Dynamics Simulations", 2021 MRS Fall Meeting & Exhibit, Boston, MA, USA, Nov. 28 – Dec. 3, 2021.
2. "Long-range Coulomb for reactive molecular dynamics simulations", An oral presentation at the 2021 E-MRS Fall Meeting, September 20 – 23, 2021.
3. "Long-range Coulomb for reactive molecular dynamics simulations", A poster presentation at the 10th International Conference of the Asian Consortium on Computational Materials Science (ACCMS-10) held at The City University of Hong Kong, Hong Kong, July 22 – 26, 2019.
4. 1st African International Conference/Workshop on "Applications of Nanotechnology to Energy, Health and Environment", held at the University of Nigeria, Nsukka on March 23 -29, 2014.
5. The 6th International Seminar on "Theoretical Physics and National Development (ISOPAND)" held at the National Mathematical centre Abuja on July 17 – 18, 2013

WORK/TEACHING EXPERIENCE

Dates	Position	Duties
2015 – 2018	**Academic content creator** @ Dreeze Academy.	▪ Develops questions and solutions on high school Maths and Physics
2015 – 2022	**Assistant Lecturer**, Department of Physics, Alex Ekwueme Federal University Ndufu-Alike, Ebonyi State, Nigeria.	▪ Taught undergraduate physics subjects, including Electronics I & II, Computational Physics, Mathematical methods of Physics and Quantum Mechanics I. ▪ First year undergraduate practical physics coordination. ▪ First year undergraduate physics experiments demonstration. ▪ Assists in the preparation, setting, and grading of exams.
2018 - 2021	**Teaching Assistant** @ Applied Physics Department, PolyU.	▪ Conducts online and in-class tutorials. ▪ Grades homework assignments and quizzes, invigilates exams. ▪ Vets teaching materials (lecture notes and Python source codes)

FINAL YEAR UNDERGRADUATE STUDENT MENTORING

Name	Degree	Date Graduated	Co-mentors	Title
Ibegbulam, Gloria Miriaku	B.Sc.	08/2017	Maduka Maduka and Eze Michael Chika	Fourth order Runge-Kutta method for numerical solution of an AC source RLC series circuit equation
Ogbonnaya, Emmanuel	B.Sc.	08/2018	Michael U. Onuu	Study of Ground-observed, Satellite-derived and Global Solar Radiation in Enugu and Owerri, Nigeria

PROFESSIONAL ORGANIZATION

- Graduate Member, African Materials Research Society (A-MRS), AUST-Chapter
- Member, Material Research Society